GNEG590V Assignment # 4

1. 3-1, 3-2, 3-5, 3-32

2. Host A and host B are communicating using the ARQ protocol in data link layer. Assume the data frame can be represented by the data structure as (seq.#, data). For example, $(0, A_i)$ means a data frame from Host A with sequence number 0, and the payload contains the ith packet from the network layer of A.

   The Ack frame can be represented by (Ack = seq. #). For example (Ack = 0) means this is an acknowledgement frame for data frame with sequence number 0.

   Assuem A sends 6 data frames to B. Following the example in page 26 of the slides of Ch. 3, sketch the conversation between host A and host B for the following cases

   (a) The 2nd data frame is lost during transmission from A to B.

   (b) The Ack for the 3rd data frame is in error.

   (c) The 5th data frame is lost during transmission from A to B.

   (d) After the 5th data frame successfully received by B, the Ack for the 5th data frame is lost.

   (e) All the other frames are successfully delivered.

3. Host A and host B are communicating using the 1-bit Sliding Window protocol in data link layer. Assume the frame can be represented by the data structure as (seq.#, Ack.#, data). For example, $(0, 1, A_i)$ means a piggybacked data frame from Host A with sequence number 0, Acknolwedgement number 1, and the payload contains the ith packet from the network layer of A. Also, there are always data from the network layer.

Let's consider the situation that, in certain time instant, host A sends out frame $(0, 0, A_i)$; the frame is succefully received by B; host B replies by $(1, 0, B_j)$.

(a) If there is no error in channel, predict the next frame that is going to be sent out by Host A and Host B, respectively.

(b) If the reply from B, $(1, 0, B_j)$, is lost during transmission, and assume the timer at Host B is long enough that it doesn't expire during the considered time, predict the next frame that is going to be sent out by Host A and Host B, respectively.